

Practice problem set

E7 Final exam

Spring 2026
« version 3 »

1) Week 11: Numerical differentiation

1.1) The distance x of a runner from a fixed point is measured in meters at intervals of half a second. The data obtained are:

t	x
0.0	0.0
0.5	3.65
1.0	6.80
1.5	9.90
2.0	12.15

Use forward, backward, and central differences to approximate the runner's velocity at times $t = 1.0$ s.

1.2) Suppose that you take samples of sea level heights at a rate of once every two hours, and you use the backward differences scheme to estimate the rate of change.

- Compute a bound on the size of the error incurred if sea level is sinusoidal with an amplitude of 3 feet and a period of 12 hours, meaning that it is generated by the function $f(t) = 3\sin(\frac{2\pi}{12}t)$, with t measured in hours.
- How often should you take measurements if you wish the backward differencing error to not exceed 0.1 ft/hr?
- Repeat using the central differencing scheme.

2) Week 11: Numerical integration

2.1) Suppose we have taken the following samples of a function $f(x)$.

x	$f(x)$
$-\frac{\pi}{2}$	1
0	0
$\frac{\pi}{2}$	1

Use this data to estimate

$$\int_{-\pi/2}^{\pi/2} f(x) dx$$

- with the composite trapezoidal rule
- with the composite Simpson's 1/3 rule

2.2) Use the following table of samples of $f(x)$ to approximate $\int_0^2 f(x)dx$ using composite Simpson's 1/3 rule.

x	$f(x)$
0.0	-2.5
0.5	3.0
1.0	2.5
1.5	2.0
2.0	-4.5

2.3) We wish to approximate the integral:

$$\int_0^3 x^4 dx$$

Find a bound on the approximation error of the following two techniques:

- a) Composite trapezoidal with $h = 0.2$.
- b) Composite Simpson's 1/3 rule with $h = 0.1$.

2.4) We wish to approximate the integral of $f(x) = cx^2 + 3x + 4$ over the interval $[-6, 6]$ with a step size of $h = 0.5$. Find the maximum value of c such that the error is guaranteed not to exceed 1 when the approximation is made with

- a) the composite trapezoidal rule, or
- b) the composite Simpson's rule.

2.5) We wish to approximate the integral of $f(x) = e^{-x} \sin(x)$ on the interval $[0, 3]$ using a segment size of $h = 0.1$. Find an upper bound on the error made by

- a) the composite trapezoidal rule
- b) the composite Simpson's rule. (Note: Your estimate of M_4 does not have to be exact).

2.6) For which of the following functions $f(x)$ is the estimate of

$$\int_0^{10} f(x) dx$$

using composite Simpson's rule with $h = 1$ guaranteed to produce an error less or equal to 0.5?

- a) $f_1(t) = 3 \sin(t)$
- b) $f_2(t) = e^{t/10}$
- c) $f_3(t) = t^5$
- d) $f_4(t) = 2t$

3) Week 12: Bisection and Newton-Raphson

3.1) Under what conditions might the bisection method fail to find a root of $f(x)$? What about Newton-Raphson?

3.2) By what factor does the interval that contains the root decrease in each iteration of the bisection method?

3.3) Carry out the first three iterations of the Newton-Raphson method for the function $f(x) = x^2 - 4$ with initial value $x_0 = 6$.

3.4) How many iterations will be taken by the bisection method to find a root of the function $f(x) = x^2 - 11$ if the initial interval is $[0, 5]$ and the tolerance is 0.5?

3.5) You wish to use the bisection method to find a root of $f(x) = x^2 - 3x + 1$ to within a tolerance of 10^{-3} .

- a) Suggest a viable initial condition for the iteration.
- b) Compute the interval after two iterations.
- c) How many iterations will it take for the algorithm to converge given your suggested initial condition?

3.6) You wish to use the bisection method to find a root of $f(x) = e^{-x} \cos(x)$ in the interval $[0, 3]$.

- a) Is $[0, 3]$ a viable initial interval for the iteration? If not, suggest another.
- b) Create a table with the first three iterations of the method.
- c) How many iterations are needed to find a root to within a tolerance of 10^{-5} ?

3.7) You wish to find a root of the function $f(x) = e^{-x} - x$, defined on the real line.

- a) Will Newton-Raphson converge to the root regardless of the initial condition? Why or why not?
- b) Carry out the first two iterations of Newton-Raphson, starting from $x_0 = 1$.

3.8) You wish to find a root of the function $f(x) = x^3 - 3x + 1$ using the Newton-Raphson method. Give an initial condition for which the approach is guaranteed to fail.

3.9) You wish to use the bisection method to find a root of the function:

$$f(x) = (x - 2.5)(x + 1)(x - 1)$$

Which of the following options are valid initial intervals?

- ☐ $[-1, 1]$
- ☐ $[0, 1]$
- ☐ $[0, 2]$
- ☐ $[0, 3]$

4) Week 12: Fixed points

4.1) We wish to find a root of the function

$$f(x) = x^2 - \cos x$$

a) Which of the following functions g have a fixed point that is a root of f ?

☐ $g_1(x) = x + x^2 - \cos x$

☐ $g_2(x) = \sqrt{\cos x}$

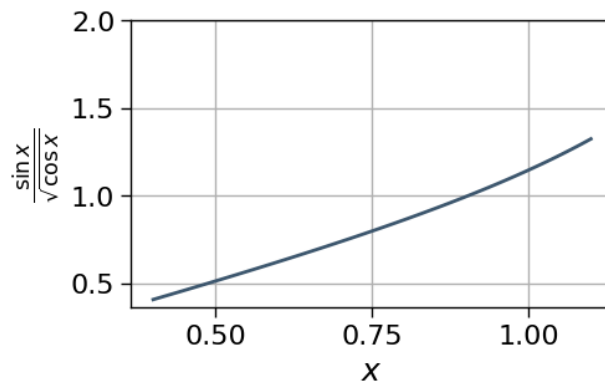
☐ $g_3(x) = x - \cos x$

b) Suppose that we know that a root exists in the interval $[0.5, 1]$. For which functions g is the fixed point iteration guaranteed to converge when initialized in that interval? (the plot of $\frac{\sin x}{\sqrt{\cos x}}$ is shown for reference).

☐ $g_1(x) = x + x^2 - \cos x$

☐ $g_2(x) = \sqrt{\cos x}$

☐ $g_3(x) = x - \cos x$



4.2) You wish to find a root of the function

$$f(x) = x^3 - 2x + 1$$

using FPI with the auxiliary function

$$g(x) = \frac{x^3 + 1}{2}$$

a) Prove that all of the fixed points of $g(x)$ are roots of $f(x)$.

b) Supposing we know that a fixed point of g exists in the interval $[0, 0.7]$, prove that FPI will converge for any initial condition in that interval.

c) Create a table with the first two iterations of the fixed point iteration starting from $x_0 = 0$

4.3) You wish to use a fixed-point iteration to find a root of the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$

$$f(x_1, x_2) = \left(-x_1 + \frac{x_2}{2} + 1, x_1 x_2 - 1 \right)$$

You consider the auxiliary function $g : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ (with $x_1 \neq 0$)

$$g(x_1, x_2) = \left(\frac{x_2}{2} + 1, \frac{1}{x_1} \right)$$

- a) Is this a valid choice, in the sense that the fixed points of g are roots of f ?
- b) Find a region of $\mathbb{R}^2$ where the fixed point iteration is guaranteed to converge to a solution, provided one exists in that region.
- c) Pick an initial condition in the region you just found and carry out the first two iterations of FPI.

4.4) Consider the function

$$g(x) = 4x^3 + x^2$$

- a) What are the fixed points of $g(x)$?
- b) Create a table with the first three iterations of FPI with initial value $x_0 = -0.5$.
- c) Find an interval for which it is guaranteed that if you start FPI in that interval, it will converge to a positive fixed point of $g(x)$.

4.5) We wish to find a root of the function $f(x) = e^{-x} - x$, defined on the real line. Suppose that we know that a root of f exists in the interval $[0.5, 1]$. Prove that $g(x) = e^{-x}$ is a good proxy function for finding this root using a fixed point iteration in the sense that a) a fixed point of $g(x)$ is a root of $f(x)$ and b) the fixed point iteration is guaranteed to converge.

5) Week 13: Matrix operations

5.1) For the matrices $A \in \mathbb{R}^{5 \times 2}$ and $B \in \mathbb{R}^{2 \times 3}$, is the matrix multiplication AB a valid operation? If so, what is the shape of AB ?

5.2) What spaces do A , x , and b belong to?

$$A = \begin{bmatrix} 0 & 1 & 5 \\ 0 & 0 & 0 \end{bmatrix}, \quad b = (8, 0), \quad x = \begin{bmatrix} 0 & 3 & 1 \end{bmatrix}$$

- ☐ $A \in \mathbb{R}^{3 \times 2}, b \in \mathbb{R}^2, x \in \mathbb{R}^3$
- ☐ $A \in \mathbb{R}^{2 \times 3}, b \in \mathbb{R}^3, x \in \mathbb{R}^2$
- ☐ $A \in \mathbb{R}^{2 \times 3}, b \in \mathbb{R}^2, x \in \mathbb{R}^3$
- ☐ $A \in \mathbb{R}^{2 \times 3}, b \in \mathbb{R}^2, x \in \mathbb{R}^2$

5.3) Given the matrix $A \in \mathbb{R}^{19 \times 23}$, what space of matrices could B belong to in order for the product BA to be well-defined? Choose all that apply.

- ☐ $\mathbb{R}^{23 \times 19}$
- ☐ $\mathbb{R}^{23 \times 23}$
- ☐ $\mathbb{R}^{46 \times 23}$
- ☐ $\mathbb{R}^{19 \times 23}$

5.4) Given the following matrices G and H :

$$G = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, H = \begin{bmatrix} 2 & 3 & 4 \end{bmatrix}$$

- a) What is the value of HG ?
- b) What is the dimension of GH ?

5.5) Given matrices $A \in \mathbb{R}^{2 \times 4}$ and $B \in \mathbb{R}^{\alpha \times \beta}$

- a) What should α be so that AB is well-defined?
- b) What should β be so that BA is well-defined?

6) Week 13: 3D Rotations and Projections

6.1) Let $p = (1, 2, 1)$ represent the position of a penny in 3D space relative to the (x, y, z) coordinate frame. Find the position of the penny in the coordinate frame (x'', y'') obtained by applying the following transformations to (x, y, z) :

- 1. Rotate (x, y, z) by 30° about the x axis. This produces the (x', y', z') coordinate frame.
- 2. Perform a projection of the (x', y', z') coordinate frame onto (x'', y'') in which the z' coordinate becomes the y'' coordinate ($z' \rightarrow y''$), and the y' coordinate becomes the x'' coordinate ($y' \rightarrow x''$).

6.2) Let $p = (p_x, p_y, p_z)$ represent a point in 3D space. Let

$$Z = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a matrix used to rotate the coordinate frame by $\beta = 45^\circ$ around the z axis. Which of the following is the correct formula for determining p 's coordinates with respect to the rotated coordinate frame?

- ☐ pZ
- ☐ Zp
- ☐ $p + Z$

7) Week 13: Systems of linear equations

7.1) Express the following system of equations in matrix form.

$$\begin{aligned} 1 - x_2 &= 3x_3 \\ x_1 &= 7 - x_2 \\ x_1 + x_2 + x_3 &= 0 \end{aligned}$$

7.2) A system of linear equations $Ax = b$ is known to have infinitely many solutions. What can be said about the rank of A ?

7.3) Express the following system of equations in matrix form.

$$\begin{aligned}x_1 + x_2 &= 1 \\x_1 + x_3 &= 2 \\x_2 + x_3 &= 3 \\2x_1 + 2x_2 + 2x_3 &= 4\end{aligned}$$

7.4) A system of linear equations is given in matrix form $Ax = b$, with $A \in \mathbb{R}^{2 \times 5}$.

- What is the dimension of x ?
- What is the dimension of b ?
- How many solutions are there to the system of equations if both $\text{rank}(A)$ and $\text{rank}(A|b)$ equal 2?

7.5) The rank of A is 3 and the rank of $A|b$ is 4. How many solutions are there to this system of equations $Ax = b$?

7.6) You are given the matrix equation $Ax = b$, where b is not in the span of A . How many possible x satisfy this equation?

8) Week 14: Initial value problems

8.1) Consider the following ordinary differential equation:

$$y'' + \alpha y'y + y = t$$

- With $\alpha = 1$, is the ODE *linear* or *nonlinear*?
- With $\alpha = 0$, is the ODE *linear* or *nonlinear*?
- With $\alpha = 1$, what is the order of the ODE?
- With $\alpha = 0$, what is the order of the ODE?
- Express the ODE in *vector form* (keeping α as a variable).
- Use Euler's method to estimate the value of $y(1)$, starting from $y(0) = 2$, $y'(0) = 1$, and with a step size of $h = 1$. Express the solution as a function of α .

8.2) Rewrite the following second-order linear ODE into a system of first-order ODEs.

$$y'' - y' + y = 0$$

- Is this a linear ODE or a non-linear ODE?
- Write the ODE in matrix form.
- With initial condition $y(0) = 1$, $y'(0) = 1$ and step size $h = 1$, perform one step of Euler's method and estimate $y(1)$.
- Perform one step of the midpoint method.
- Perform one step of Heun's method.

8.3) Consider the following ODE:

$$y''' + 3t^2 y' - 2y = e^{-t}$$

- a) What is its order?
- b) Is it linear or nonlinear?
- c) Express the ODE in vector form. If linear, express it in matrix form.
- d) Compute the first iteration of the Heun's method with initial condition $y = 0$, $y' = 1$, $y'' = 0$, and time step $h = 1$.

8.4) The Duffing equation is used to model certain damped and driven oscillators. The Duffing equation is given by:

$$\beta y^3 + y'' + \delta y' + \alpha y = \gamma \cos(\omega t)$$

where δ is the damping coefficient, α and β are the linear and non-linear stiffness coefficients, respectively, and $\gamma \cos(\omega t)$ represents the driving force with amplitude γ and frequency ω .

- a) What is its order?
- b) Is it linear or nonlinear?
- c) Express the ODE in vector form. If linear, express it in matrix form.

8.5) Consider the following ODE:

$$y'' - \mu(1 - y^2)y' + y = 0$$

- a) What is its order?
- b) Is it linear or nonlinear?
- c) Express the ODE in vector form. If linear, express it in matrix form.
- d) Compute the first iteration of the midpoint method with initial condition $y = 0$, $y' = 1$, time step $h = 0.5$, and $\mu = 1$.

8.6) Consider the following ODE:

$$y'' - 4y' + 4y = 10$$

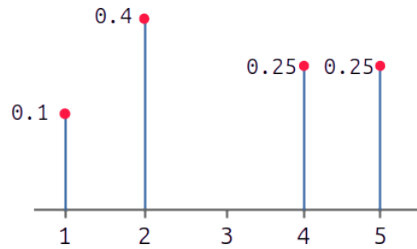
- a) What is its order?
- b) Is it linear or nonlinear?
- c) Express the ODE in vector form. If linear, express it in matrix form.
- d) Compute the first two iterations of Euler's method with initial condition $y = 0$, $y' = 0$ and time step $h = 0.1$.

9) Week 15: Randomness

9.1) The variable X is uniformly distributed over the interval $[1, 3]$.

- Give a formula for the distribution of X .
- Compute the probability of the event $e = [\frac{3}{2}, \frac{5}{2}]$.

9.2) For the discrete distribution shown in the figure below, answer the following.



- What is its sample space?
- Calculate its mean.
- Calculate its standard deviation.
- What is the probability of the event $\{1, 2\}$?

9.3) The function $p(x) = c x^2$ is defined on the interval $\Omega = [0, 3]$.

- What must the value of c be for $p(x)$ to be a valid distribution?
- Find the mean of X (the random variable whose distribution is $p(x)$).
- What is the probability of sampling a value of X in the interval $[1, 2]$?

9.4) The function $p(x)$ is defined on the real line.

$$p(x) = \max\left(0, c - \frac{|x|}{2}\right)$$

- Draw this function for $c = 1$.
- What must c be for $p(x)$ to be a valid distribution?
- With c set as in part b), what is the probability of the event $[-c, c]$?

9.5) A Monte Carlo simulation has been run to estimate the variation of the temperature at the center of a heated plate when the temperatures at its corners are subjected to random variations. Upon running 3 executions of the code, we obtain the following dataset (expressed in degrees Celsius).

$$\mathcal{D} = \{51.7, 53.6, 52.9\}$$

- Compute the sample mean.
- Compute the sample standard deviation.
- Estimate the probability that the temperature falls between 51°C and 53°C .